



Swiss Institute of
Bioinformatics

Long Research Stay: Swiss Institute of Bioinformatics (SIB)

Elias Crum

KNoWS Presentation
May 15th, 2025

Talk Outline

fwo Long Research Stay Grant Application



Logistics and Takeaways



Highlights of Research Stay



SIB Offices Visited



Research worked on



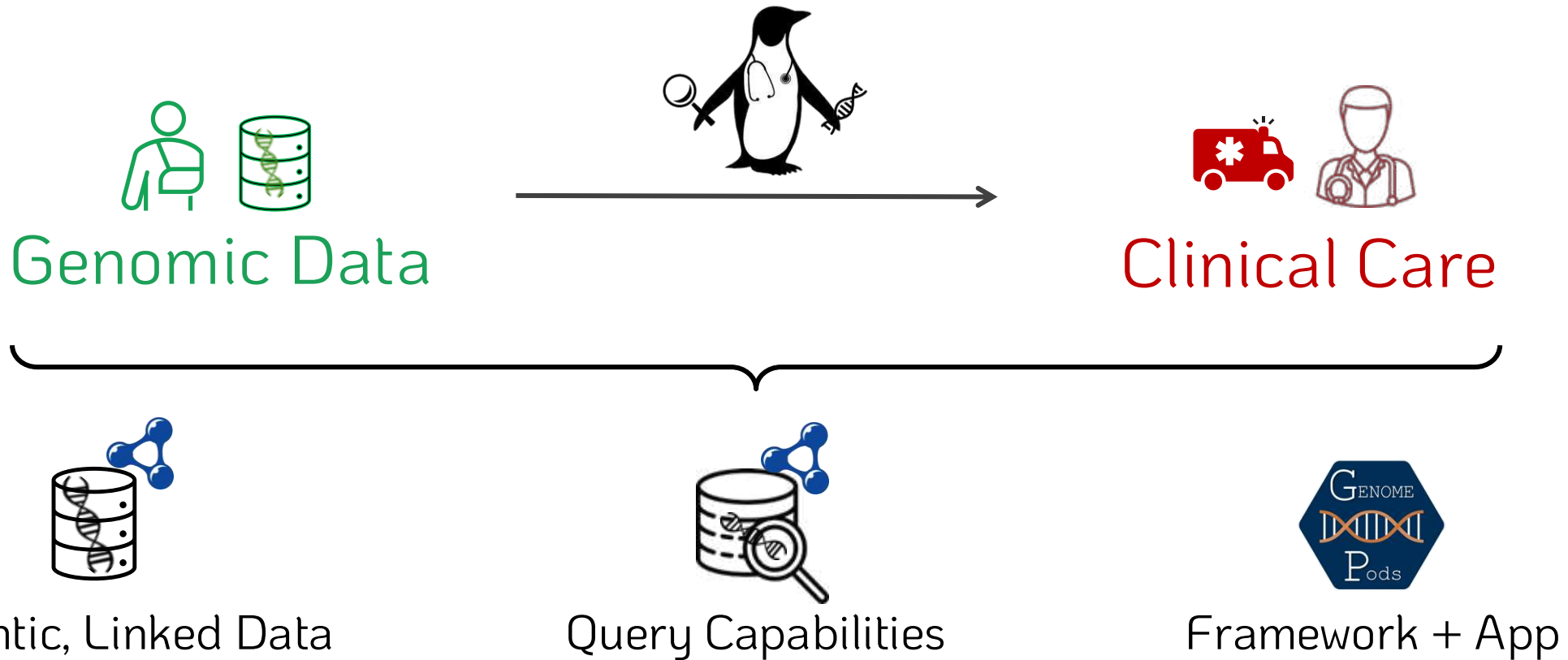
Planned Publications / Collaborations



Future aims

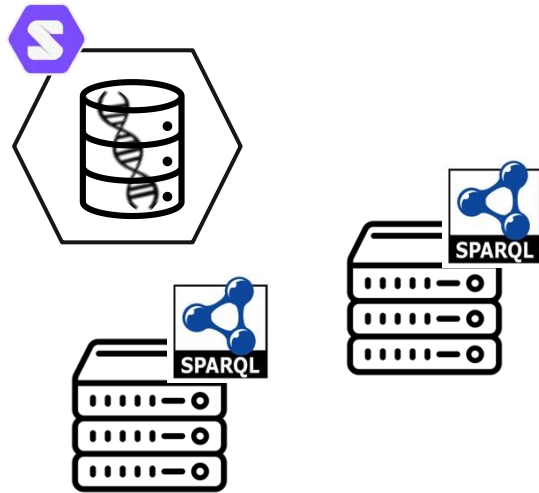
PENGQUIN:

PErsoNal Genome QUery IN healthcare and clinical practice

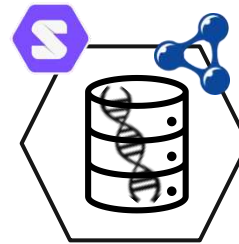


PENGQUIN Goals

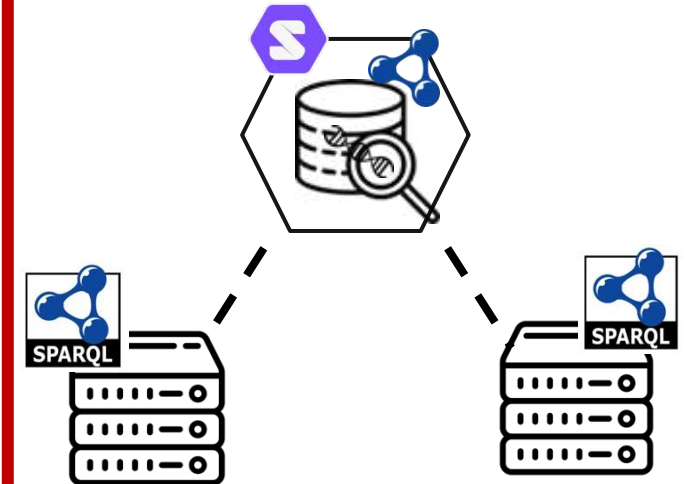
Data Storage



Genomic Data Representation



Question Answering



**Framework
& web application**



Research Stay Motivation

The Swiss Institute of Bioinformatics (SIB)



↳ Research Organization (not an Academic Institution)

- Technically my research stay was an “internship”

↳ SIB Mission

- Progress data science through in-depth knowledge of biological data, cutting-edge technologies, and interdisciplinary collaborations

Why a research stay here?

↳ Large, Public SPARQL Endpoints

↳ Swiss Personalized Health Network (SPHN)

↳ Pre-existing Collaboration – CHIST-ERA TRIPLE Project



Funding – FWO Long Research Stay Grant

Long Research Stay Application



↳ Submission due 3-months prior to stay

↳ Application Materials

- Written Application (max 5000 chars each)
 1. What is the added scientific value of the stay?
 2. Motivation for the choice of the non-Belgian host organization
 - Including 3 publications from the organization
 3. Provide a work plan, i.e. the different work packages and a detailed timetable
- 3 Personal Publications
- Letter of Recommendation from your Promoter
- Letter Invitation from host organization

Grant Application Takeaways

Hardest part is coordinating with host organization

Overall, was quite low intensity

Monetary award is suitable for ~most~ places (Switzerland was a bit of a stretch...)

I would HIGHLY recommend taking advantage of this

The Research Stay

A disappointing start ...

- ↳ Arrived in Switzerland to discover my accommodations were fraudulent (big sad face)
- ↳ After that, it was very enjoyable!



The SIB Network



SwissProt



Swiss Personalized Health Network (SPHN)



SIB Vital-IT



FRANCE

GERMANY

AUSTRIA

ITALY

Bern

SWITZERLAND

Zurich

Sankt Gallen

Zug

Chur

Fribourg

Sion

Neuchatel

Lausanne

Geneva

Basel

Lausanne



Zurich

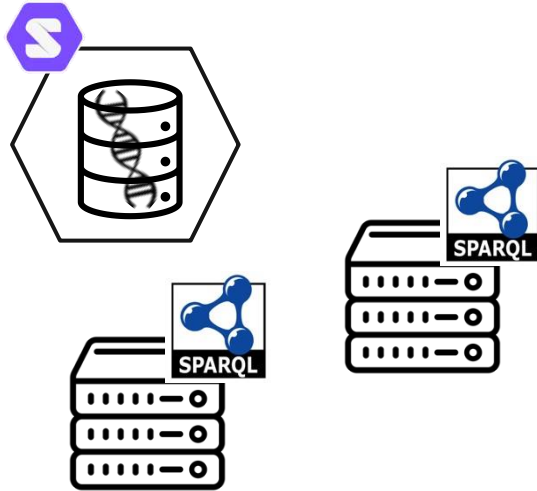


Basel

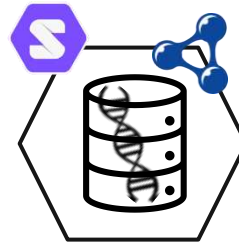


What I Worked On:

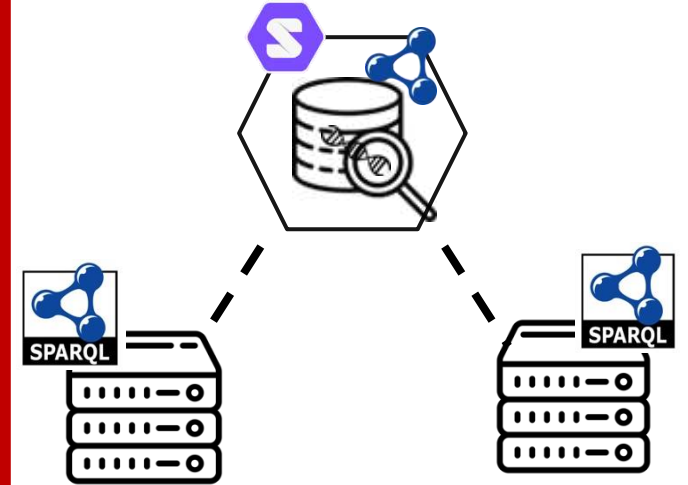
Data Storage



Genomic Data Representation



Federated Querying

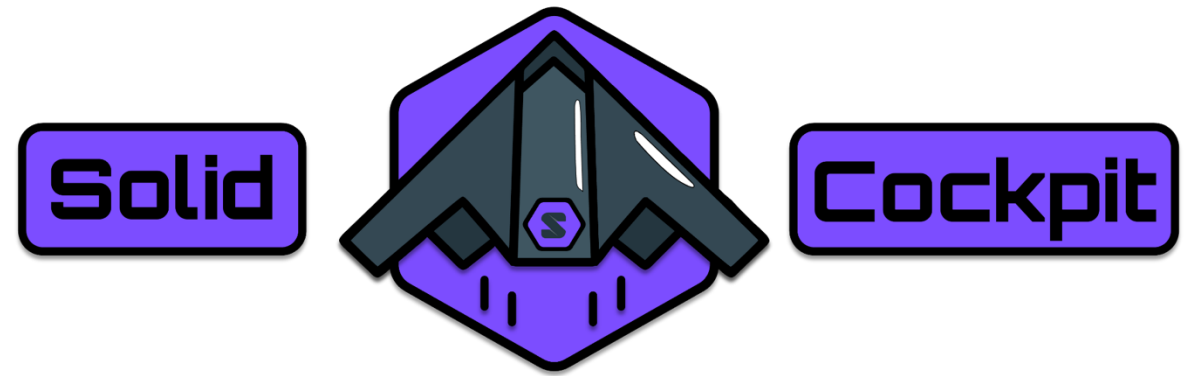


CHIST-ERA TRIIPLE Project



**Standardized documentation
and resource statistics**

Data Storage Progress



A Web Application to:

knowledgeonwebscale.github.io/solid-cockpit/

☁️ Upload Data to a Solid Pod

🗂️ Browse contents of a Solid Pod

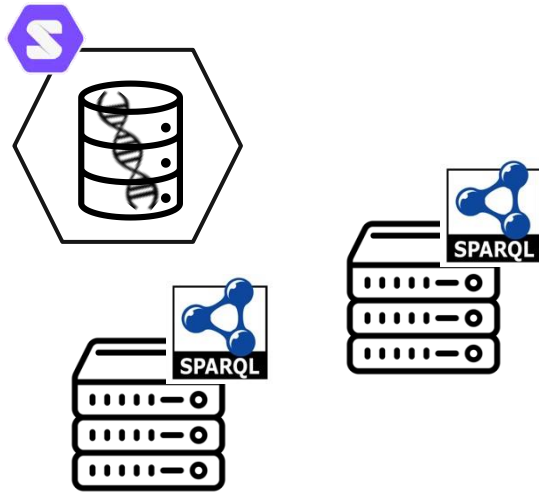
🔒 Edit Privacy of (and share) data in a Solid Pod

🔍 Query Data in Solid Pods + SPARQL Endpoints

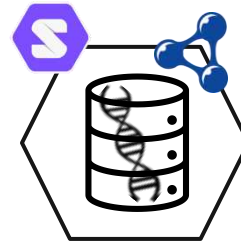
↳ Implement a query cache for storing queries and results

Semantifying Genomic Variant Data

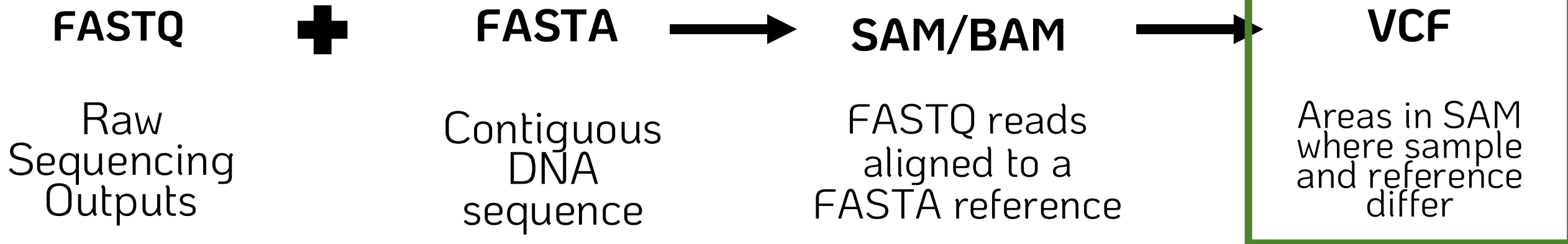
Data Storage



Genomic Data Representation



Semantic VCF – Sequencing Data Formats Intro



1	>gi 568336023 gb CM000663.2 Homo sapiens chromosome 1, GRCh	
2	CCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC	@HD VN:1.0 SO
3	ACCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAA	@SQ SN:chr20
4	TAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCT	@PG ID:TopHat
5	CTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC	lign-edit-dist 2 -
6	CAACCCCAACCCCAACCCCAACCCCAACCCCAACCCCTAACCC	20 /data/user446/m
7	CCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAAC	HWI-ST1145:74:C101
8	TAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCC	CCGTGTTTAAA
9	TCTGACCTGAGGAGAAGCTGTGCTCCGCCTTCAGAGTACCAC	C BBDCCDDCCDDDD
10	CGCTCTCGCGGTGCTCTCGGGTCTGTGCTGAGGAGAACGCA	AS:i:-15
11	CGGCGCAGGCGCAGAGAGGCGCGCCGCGCCGGCGCAGGCGCA	HWI-ST1145:74:C101
12	AGAGAGGCGCGCCGCGCCGGCGCAGGCGCAGAGAGGCGCGCC	CGTGGATCAT
13	CGCGCCGGCGCAGGCGCAGACACATGCTAGCGCGTCTGGGGTG	G DCDDDEDDDDDD
14	GCGCGCCGGCGCAGGCGCAGAGACACATGCTACCGCTCCAC	AS:i:-16
15	AGGCGCACCGCGCGCGCAGGCGCAGAGACACATGCTAGCG	
16	GCAGAGACGCAAGCCTACGGGCGGGGTTGGGGGGCGTGTG	
17	GCTGGGCGGGGGGAGGGTGGCGCCGTGCACGCGCAGAACT	
18	GGTAGAACCTCAGTAATCGAAAGACGGGATCGACCGCCCC	
19	GCTTGCTACGGTGCTGTGTCAGGGCGCCCTGTGCGGAC	
20	GAGTGGTGGCCAGCGCCCCCTGCTGGCGCCGGGCACTGCAG	

@FORJUSP02
CCGTCAATTC
+
AAAAAAAAAA

Q scores (as ASCII ch

Base=T, Q=':'=25

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
```

```
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
```

FORMAT	NA00001	NA00002	NA00003
GT:GQ:DP:HQ	0 0:48:1:51,51	0 0:48:8:51,51	1/1:43:5:.,.
GT:GQ:DP:HQ	0 0:49:3:58,50	0 1:3:5:65,3	0/0:41:3
GT:GQ:DP:HQ	1 2:21:6:23,27	2 1:2:0:18,2	2/2:35:4
GT:GQ:DP:HQ	0 0:54:7:56,60	0 0:48:4:51,51	0/0:61:2
GT:GQ:DP	0/1:35:4	0/2:17:2	1/1:40:3

```
@HD      VN:1.0  SO
@SQ      SN:chr20
@PG      ID:TopHat
lign-edit-dist 2 -
20 /data/user446/m
HWI-ST1145:74:C101
      CCGTGTTTAAA
C      BBDDCCDDCCDDDD
      AS:i:-15
HWI-ST1145:74:C101
      TGTGGATCAT
G      DCDDDEDDDDDD
      AS:i:-16
```

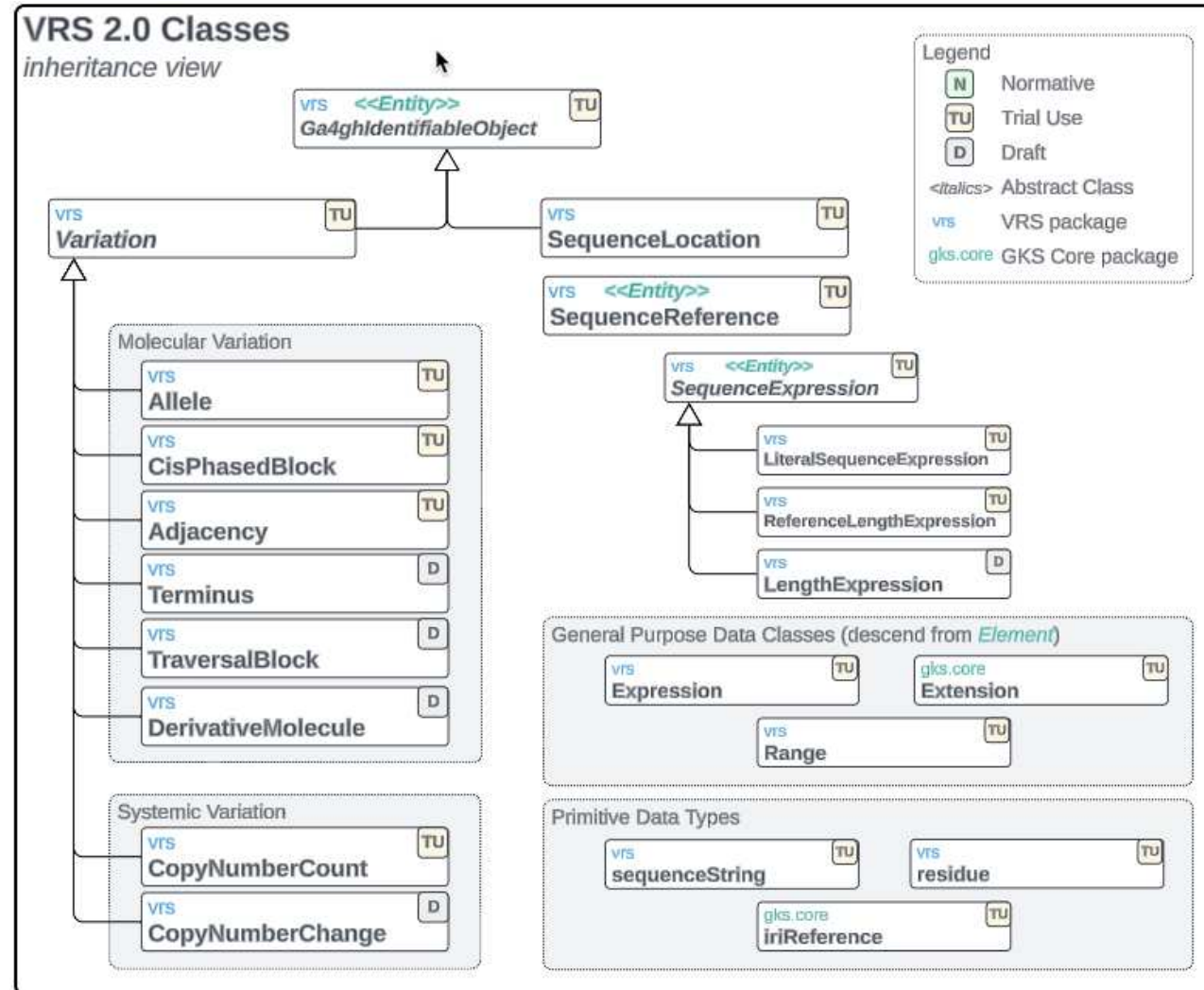
```

HWI-ST1145:74:C101DACXX:7:1204:14760:4030      16      chr20      270877      50      100M      *      0      0
      GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C      DDDDDDDDDCCDDDDDDDDDEEEEEEEFFFEFFEGHHHFGDJJIHJJIJJJJIIIGGFJJJIHIIJJJJJJIGHHFAHGFHJHFGGHHFFDD@BB
      AS:i:-11      XM:i:2      XO:i:0      XG:i:0      MD:Z:0A85G13      NM:i:2      NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699      0      chr20      271218      50      50M4700N50M      *      0
      0      GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCCTTGGGTCTCCGAGCAGAACATCCTCAAATATGACCTCTCG
accepted hits.sam

```

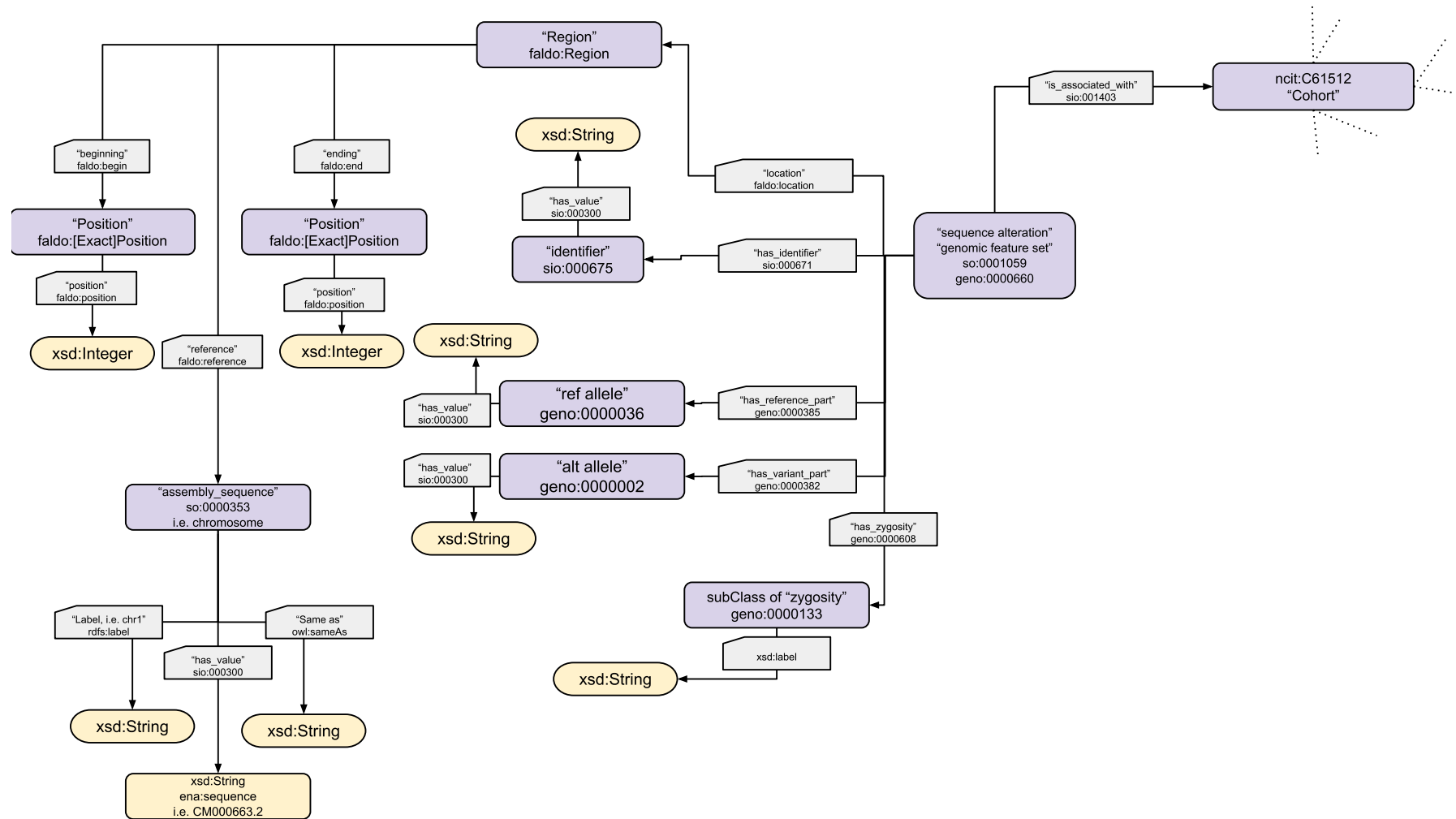
Representing VCF Semantically?

GA4GH Variant Schema:



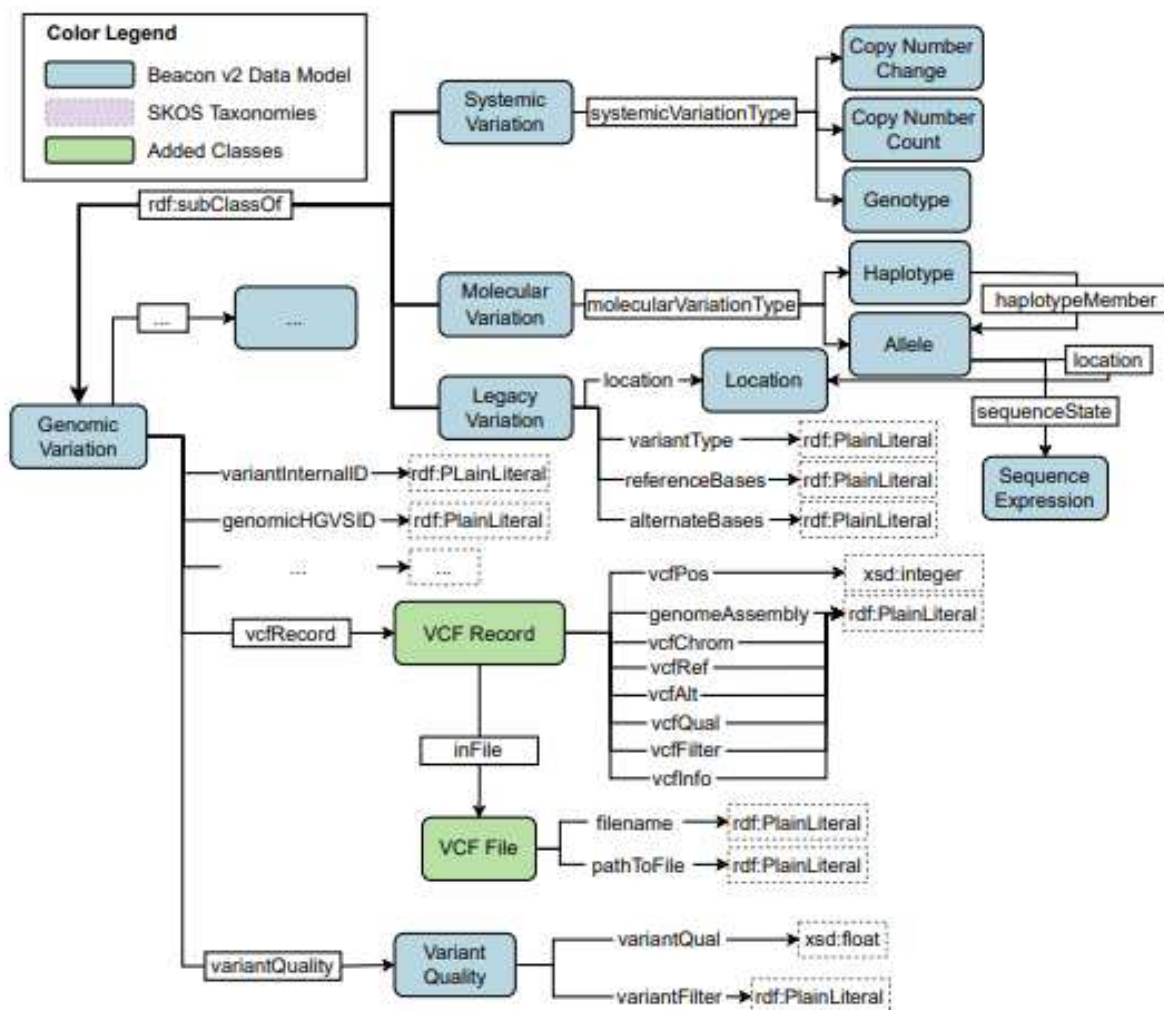
Existing Semantic VCF Resources – Semantic Beacons

Semantic Beacons Schema:



Existing Semantic VCF Resources – HERO Genomics

HERO Genomics Ontology:



Semantic VCF - Future Aims

Construct VCF Knowledge Graph

manual
conversion

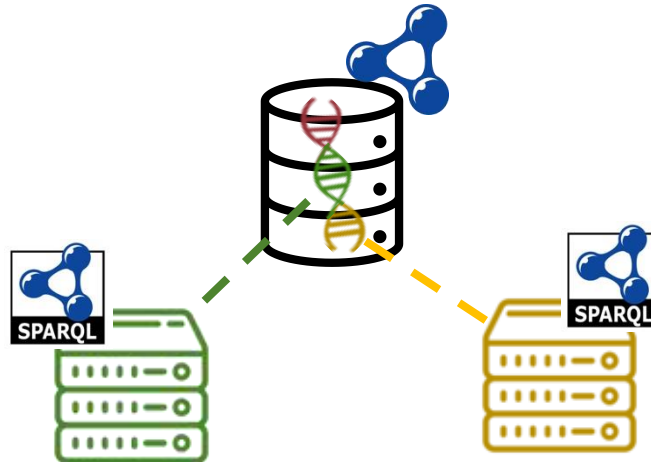


OR

RocketRML

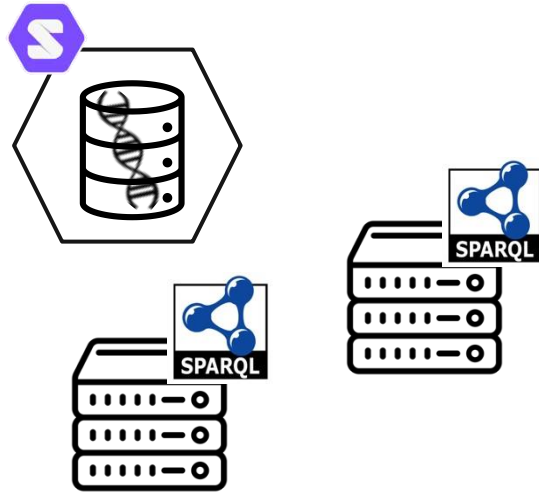


Experiment with SPARQL Querying and Data Linking

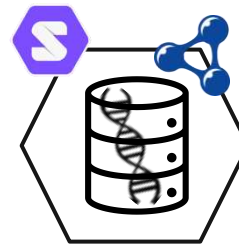


SPARQL Federated Querying

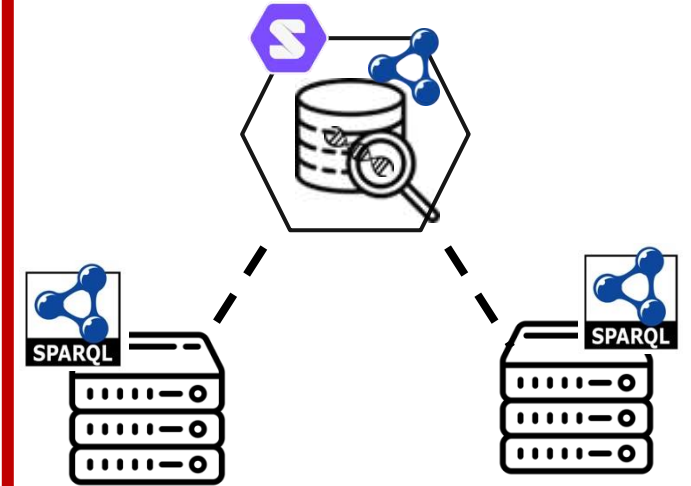
Data Storage



Genomic Data Representation



Querying



Motivation:

Federated querying for answering questions over **multiple heterogeneous data sources**

① Performance is currently *~variable~*

↳ We want to detail *WHY* performance varies so much && suggestions for improvement

② Assess federation algorithms on REAL endpoints

↳ **FedX** – uses ASK queries to inform source-selection

↳ **SPLendid** – VOID descriptions to inform source-selection

Real World Federated Query Survey - Design

SPARQL Endpoints:



“Real World” Federated Queries:



<https://sib-swiss.github.io/sparql-examples/>

SPARQL Query Engine:



<https://comunica.dev/>

Example Federated Query

Query: 027-biosodafrontend

Question: (*Bgee* / *UniProt*)

Retrieve the *genes* expressed
in the *human pancreas*

and

those *genes'* *annotations*

```
https://www.bgee.org/sparql/

1 PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
2 PREFIX up: <http://purl.uniprot.org/core/>
3 PREFIX obo: <http://purl.obolibrary.org/obo/>
4 PREFIX genex: <http://purl.org/genex#>
5 PREFIX lscr: <http://purl.org/lscr#>
6 PREFIX orth: <http://purl.org/net/orth#>
7 SELECT DISTINCT ?geneEns ?uniprot ?annotation_text {
8   {
9     SELECT ?geneEns {
10       ?geneB a orth:Gene .
11       ?geneB genex:isExpressedIn ?cond .
12       ?cond genex:hasAnatomicalEntity ?anat .
13       ?geneB lscr:xrefEnsemblGene ?geneEns .
14       ?anat rdfs:label 'pancreas' .
15       ?geneB orth:organism ?o .
16       ?o obo:RO_0002162 ?taxon2 .
17       ?taxon2 up:commonName 'human' .
18     } LIMIT 100
19   }
20 SERVICE <https://sparql.uniprot.org/sparql> {
21   ?uniprot rdfs:seeAlso / up:transcribedFrom ?geneEns .
22   ?uniprot up:annotation ?annotation .
23   ?annotation rdfs:comment ?annotation_text .
24 }
25 }
```

Example obtained from:

https://biosoda.expasy.org/build_biosodafrontend/

visualization from: <https://sib-swiss.github.io/sparql-editor/>

Real World Federated Query Survey – [Preliminary] Findings 1

Preliminary Results (endpoints and queries):

Federated querying is not "easy"

↳ Errors related to SPARQL endpoint availability

Invalid SPARQL endpoint response from <https://semantic.eea.europa.eu/sparql> (HTTP status 404)

↳ Errors caused by SPARQL endpoint-related errors

Invalid SPARQL endpoint response from <https://sparql.orthodb.org/sparql> (HTTP status 500)

↳ Errors related to SPARQL endpoint rate-limiting

Invalid SPARQL endpoint response from <https://query.wikidata.org/sparql> (HTTP status 429): Rate limit exceeded

↳ Errors related to SPARQL endpoint time-outs

Unexpected ""!"" at position 0 in state STOP

Real World Federated Query Survey – *[Preliminary] Findings 2*

Preliminary Results (algorithms):

Federated querying is not "easy"

↳ Errors within Comunica

Actor urn:comunica:default:query-operation/actors#source requires an operation with source annotation.

fetch failed

Could not retrieve <https://idsm.elixir-czech.cz/sparql/endpoint/chebi> (HTTP status 400)

FATAL ERROR: Ineffective mark-compacts near heap limit Allocation failed - JavaScript heap out of memory

↳ Errors related to algorithms

Actor urn:comunica:default:/actors#rdf-join cannot complete join with Asym-hash actor.

Planned Publications

Real-World Federated Query Survey (conference)

- ↳ SEMANTiCS 2025 (Oct 2025)
 - Long Research Paper (May 23rd)
 - Get feedback from community

Real-World Federated Query Survey (journal)

- ↳ Journal Undecided (~Winter 2025-2026)

Solid Cockpit (conference)

- ↳ ESWC 2026 (June 2026)
 - Resource Paper (Dec 2025)

VCF to RDF Implementation (conference)

- ↳ SWAT4HCLS 2026 (Feb 2026)
 - Full Paper (Dec 2025)

Planned Collaborations



Real-World Fed Query Survey

- 3 contributing authors from SIB
- Planning to present findings in Switzerland after submissions
- CHIST-ERA Project finishes in Oct 2025
- UGent promoters submitted joint grant proposal for 2026-2030



VCF to RDF Follow-Up

- Planned extension to existing SHPN RDF Schema
- Will keep in close contact

SIB Research Stay: 1 March – 30 April

Context + Thank You!



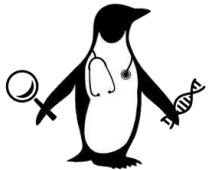
Tarcisio Mendes

Ana-Claudia Sima

Jerven Bolleman

Marco Pagni

Thank you!



More PhD specifics

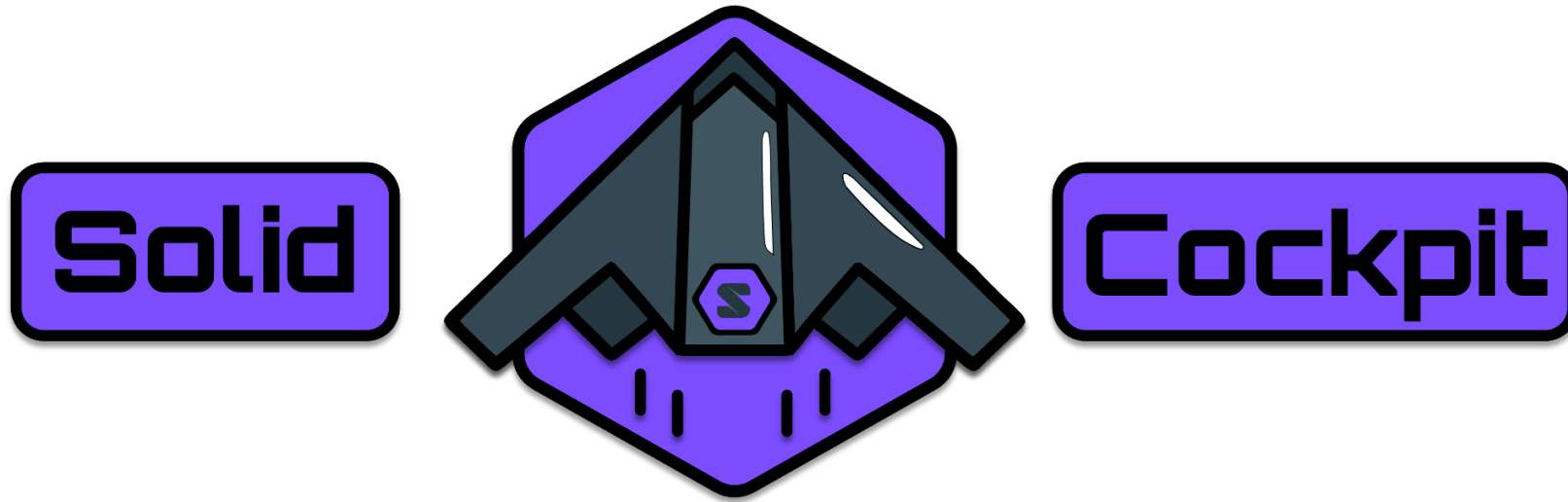


LinkedIn



GitHub

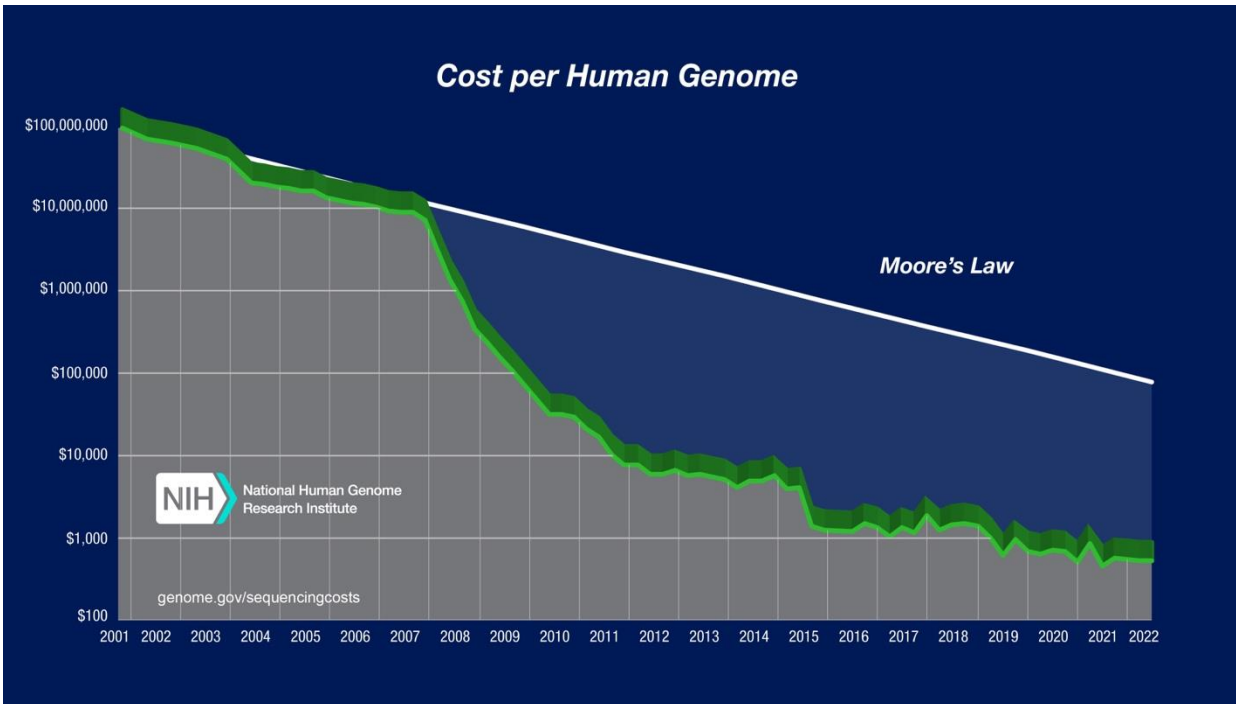
Supplemental Slides



knowledgeonwebscale.github.io/solid-cockpit/

Genome Sequencing in Health Care

Sequence Generation Costs



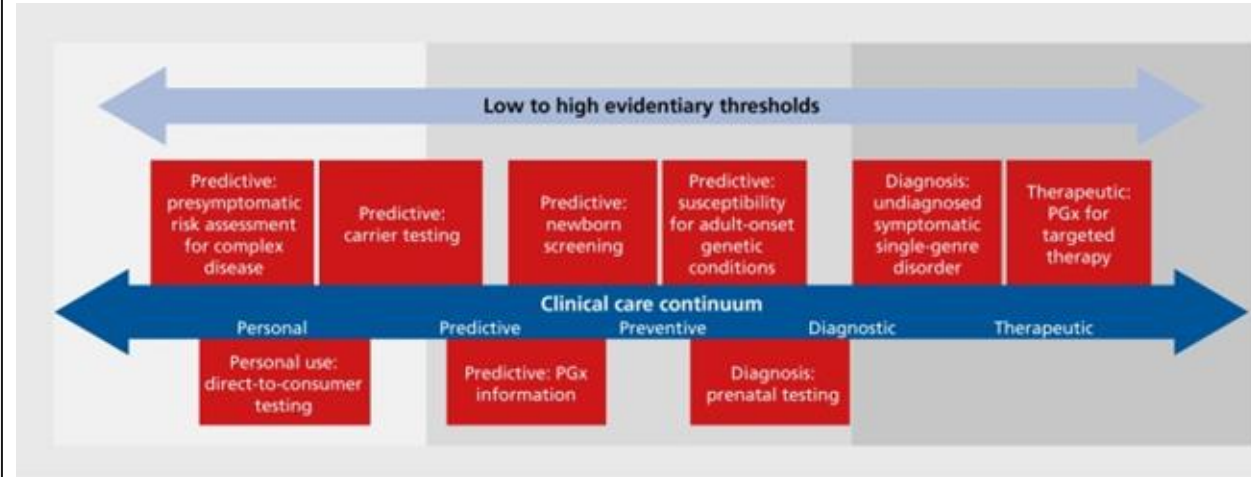
Wetterstrand KA. Accessed 12/04/24.

Currently getting close to \$100 per genome⁴

Per-base cost of sequencing has been dropping by half every ~5 months

Per-base cost of sequence data storage is dropping by half every ~14 months

Use-cases



10.31887/DCNS.2016.18.3/jkrier

Examples include:

- Medication prescription (pharmacogenomics)
- Genetic disease screening
- Cancer diagnosis & treatment

Variant Call Format (VCF)

- tab delimited .txt file
- Rows represent individual mutations in a sequence (when compared to reference)
- Columns store information about that mutation

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT
--------	-----	----	-----	-----	------	--------	------	--------



```
##fileformat=VCFv4.1
##fileDate=20140930
##source=23andme2vcf.pl https://github.com/arrogantrobot/23
##reference=file:///23andme_v3_hg19_ref.txt.gz
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT GEN
chr1 82154 rs4477212 a . . . . GT 0/0
chr1 752566 rs3094315 g A . . . . GT 1/1
chr1 752721 rs3131972 A G . . . . GT 1/1
chr1 798959 rs11240777 g . . . . GT 0/0
chr1 800007 rs6681049 T C . . . . GT 1/1
chr1 838555 rs4970383 c . . . . GT 0/0
chr1 846808 rs4475691 C . . . . GT 0/0
chr1 854250 rs7537756 A . . . . GT 0/0
chr1 861808 rs13302982 A G . . . . GT 1/1
chr1 873558 rs1110052 G T . . . . GT 1/1
chr1 882033 rs2272756 G A . . . . GT 0/1
chr1 888659 rs3748597 T C . . . . GT 1/1
chr1 891945 rs13303106 A G . . . . GT 0/1
```

Personal Genome Sequence Data

Human genome data is big...

~3.2 BILLION base pairs ----> ~1 GB

Sequencing output file ----> ~200 GB

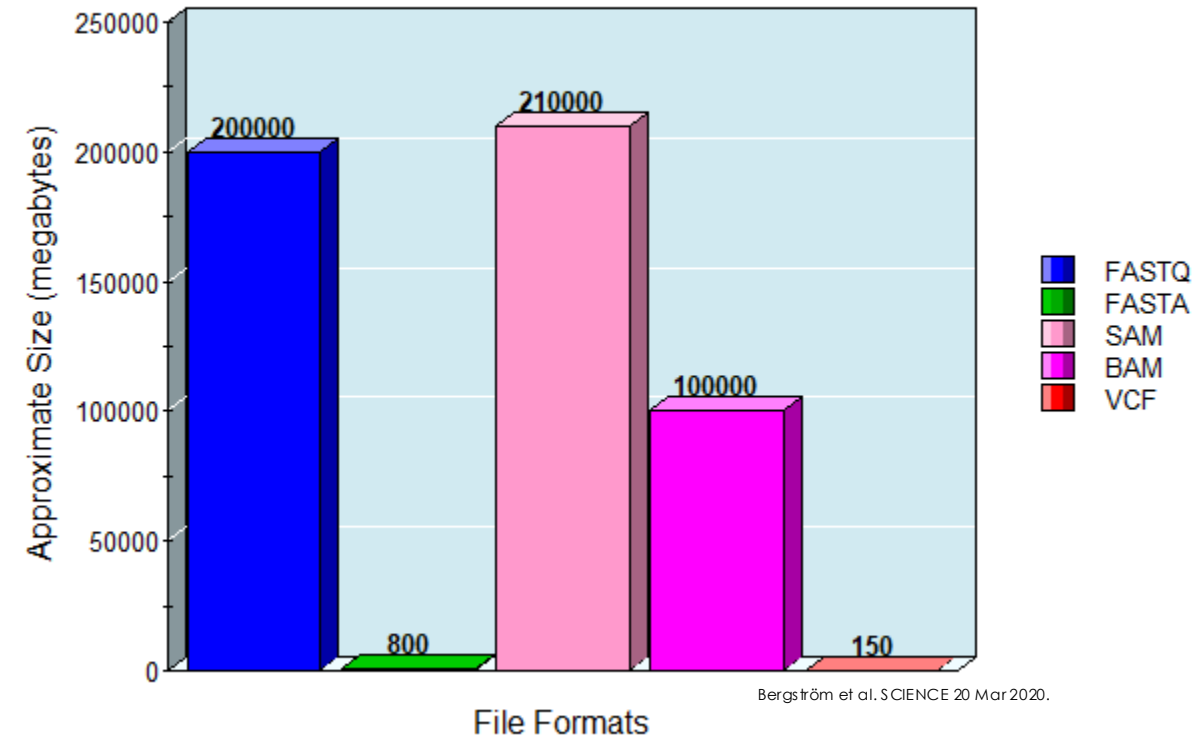
Variant Call Format (VCF) file ----> ~200 MB

- Individual humans exhibit ~3 million mutations
- VCF file **records only those differences**
- Flat file

VCF file semantic representation

- ~ >**27 MILLION** Triples when represented as RDF
- Ontology for conversion
- Header Dictionary Triples (for compression)

Sequencing File Sizes



Bergström et al. SCIENCE 20 Mar 2020.

Scaling Genome Data Usage

Research genomic data sharing initiatives:



State-of-the-art for **clinical** genomic data sharing?

GDPR (EU) / HIPPA (US)
EU Health Data Spaces → Not much...

Experimental VCF Data

Primary dataset from publicly available repositories (>100 genomes)

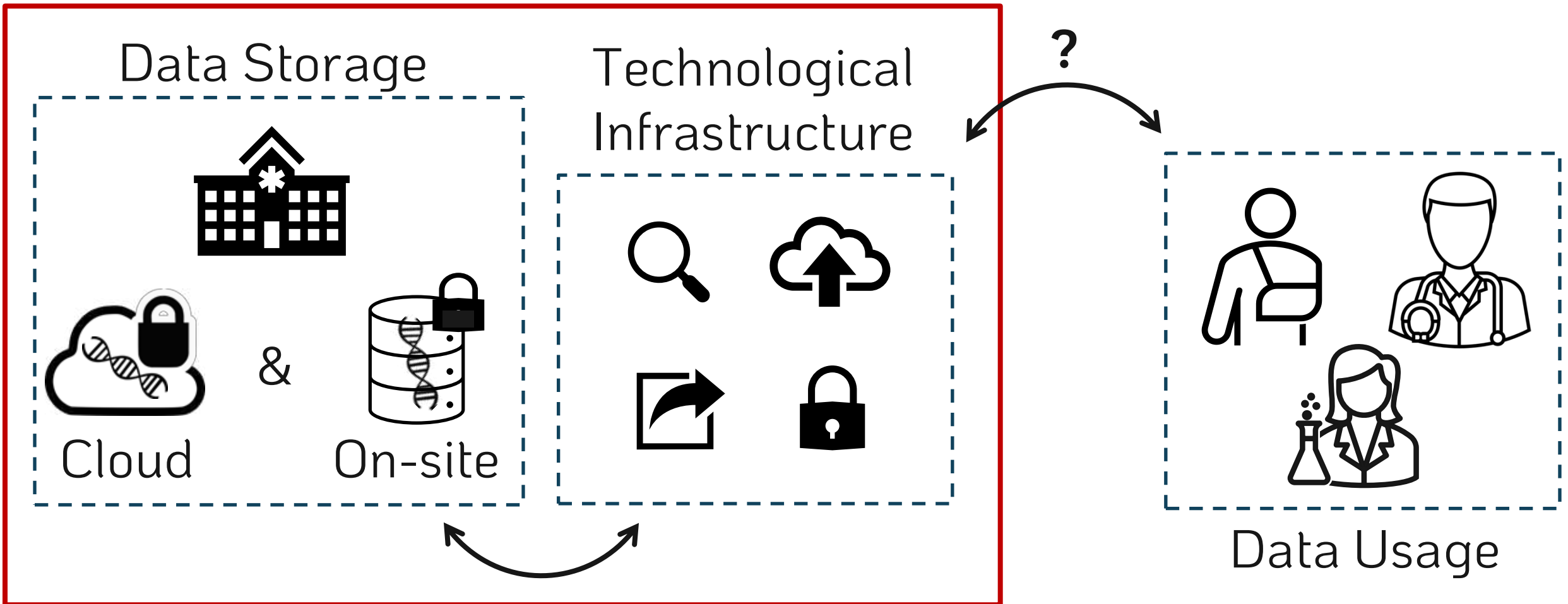
- Personal Genome Project (https://my.pgp-hms.org/public_genetic_data)
- NIST's Genome in a Bottle public dataset (<https://data.nist.gov/od/id/mds2-2336>)
- Illumina Platinum Genomes repository (<https://emea.illumina.com/platinumgenomes.html>)

If more are needed (>1000 [research grade] genomes):

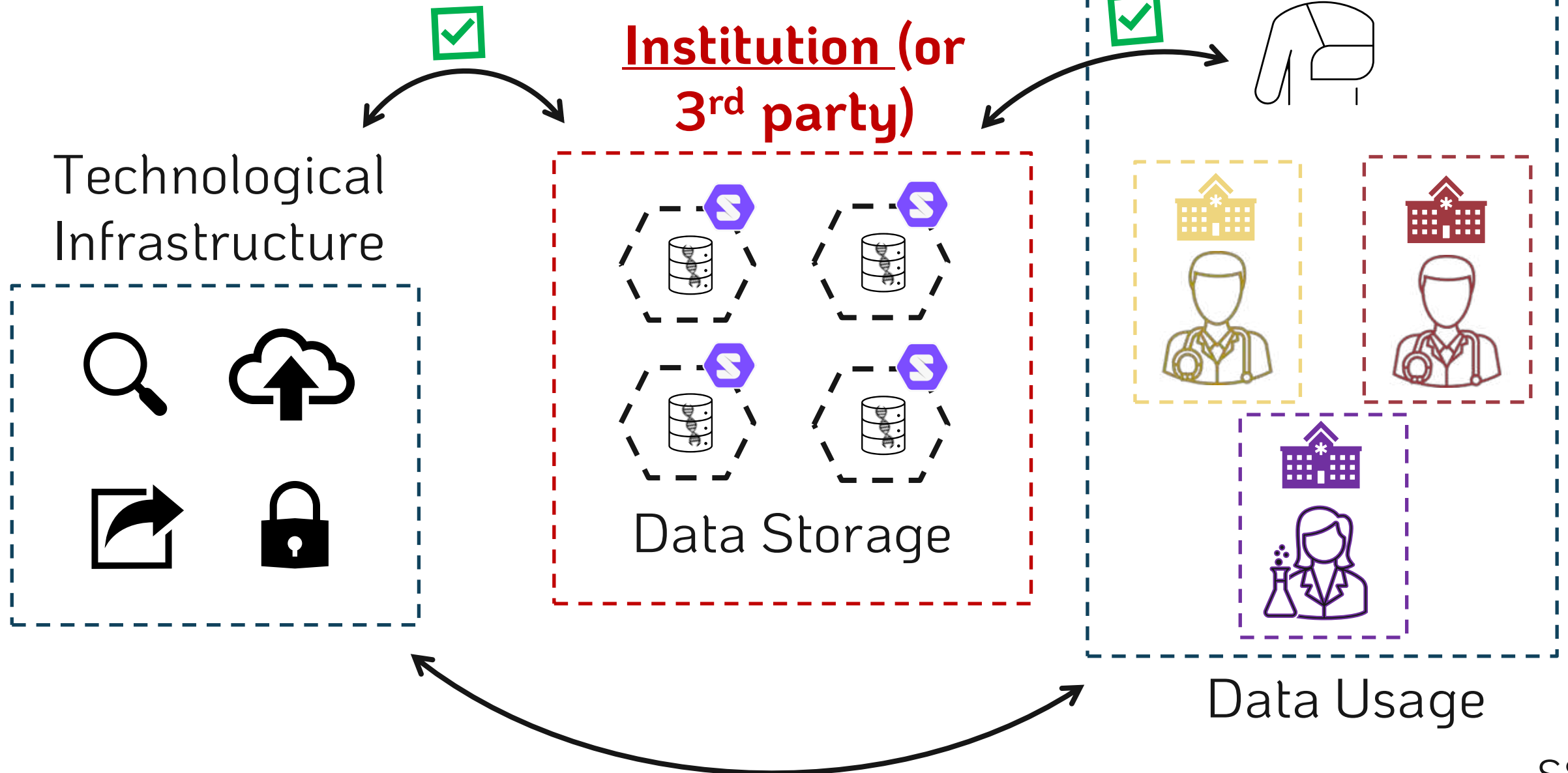
- 1000 genomes project (<https://www.internationalgenome.org/>)
- NCBI's ClinVar (<https://www.ncbi.nlm.nih.gov/clinvar/>)

Semantic Genome Data?

Institution-centric



A [Hybrid] Solid Solution





- Does not solve ALL problems, but offers an environment where improvements are possible
 - ✓ Easier sharing (web-facilitated)
 - ✓ Granular privacy controls
 - ✓ Direct patient data access
 - ✓ Data representation flexibility (RDF)
 - ✓ Possibility of data-linking and querying

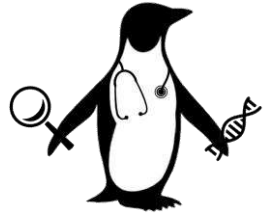
Project Goals



Create a proof-of-concept
storage **FRAMEWORK**
and accompanying **Web Application**

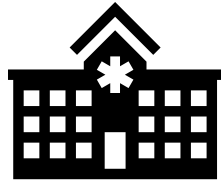
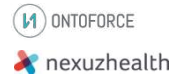
While **exploring representation + querying limits**

Envisioned Product Evolution



PhD End

PhD
Deliverable



Short Term

Start-up / Product
development
+
Hospital implementation



Medical record



Medium Term

Integrate with EHR
+
Increase application
domains

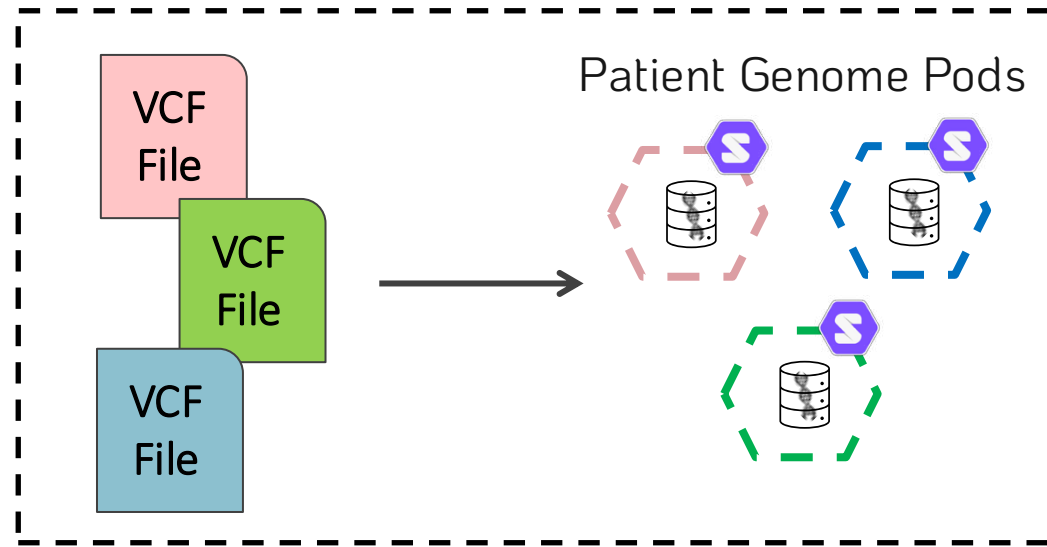


Long Term

Increase 3rd party
application support
+
Scale to Europe and
beyond

Workflow Overview

⚙️ Data Storage



Data Storage



Solid can be implemented for genomic data storage.



Solid pod instances → Community Solid Server¹

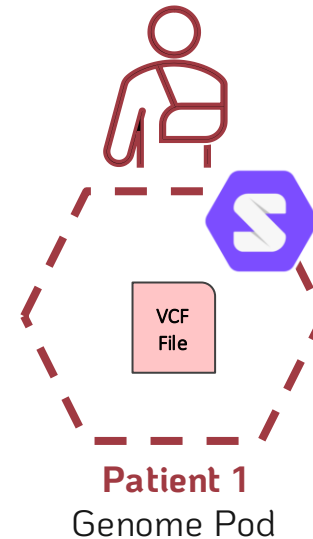


VCF Genomic Data → Illumina Platinum Genomes²

Patient 1
Genome Data

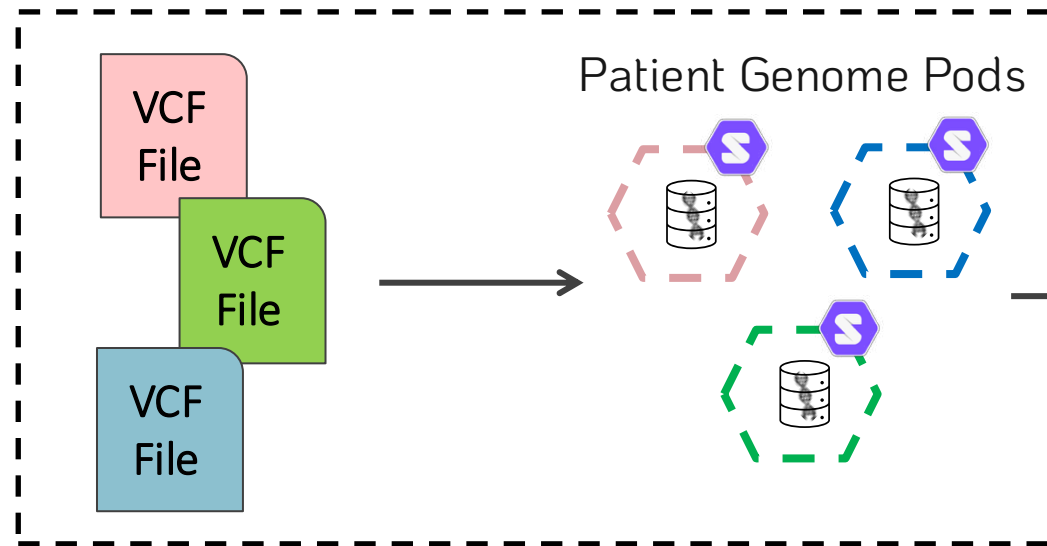


Data Ingestion

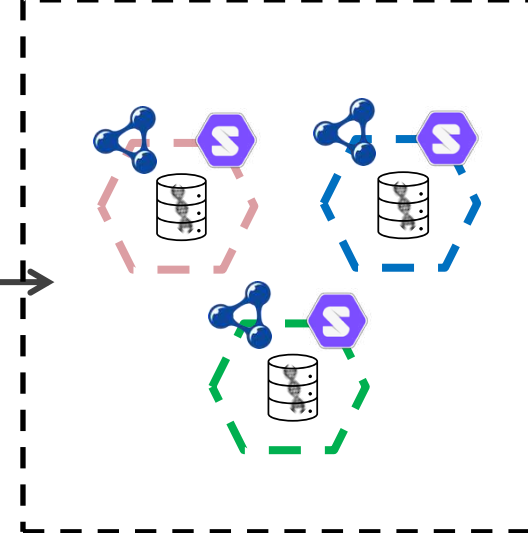


Workflow Overview

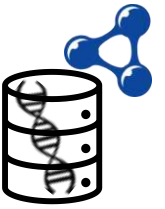
Data Storage



Representation



Genome Data in RDF

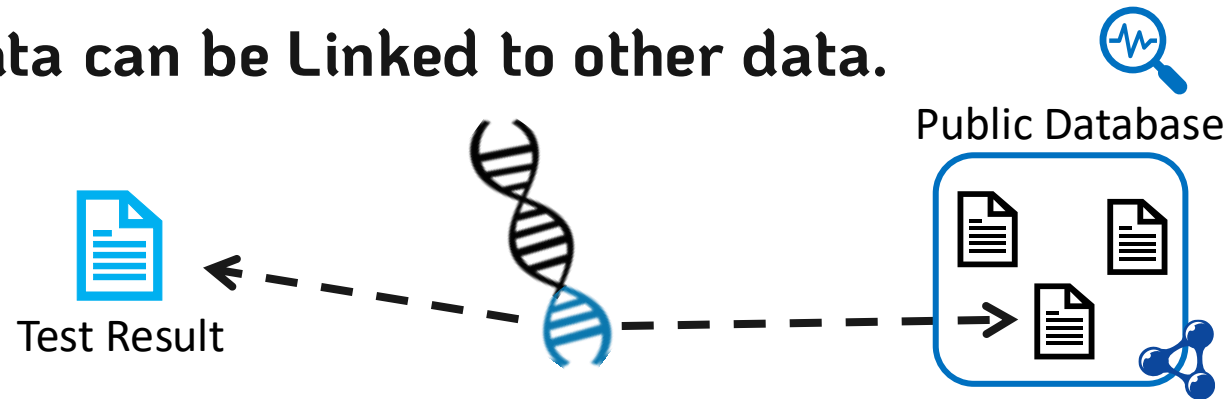


VCF genomic data can be represented as RDF.



 VCF to RDF Vocabularies exist $\longrightarrow$ SPHN Semantic Interoperability Framework⁴
+ FHIR HL7 Format⁵

VCF genomic data can be Linked to other data.



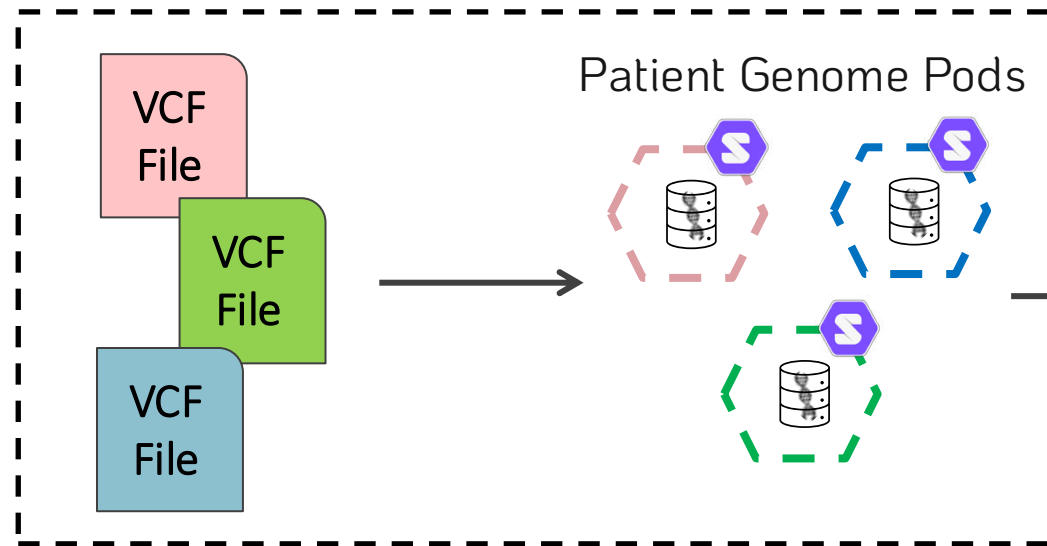
How to automate this?

What vocabularies could be used for the linkages?

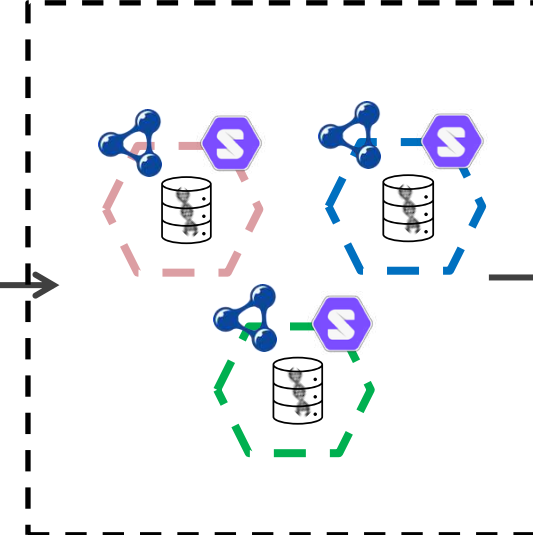


Workflow Overview

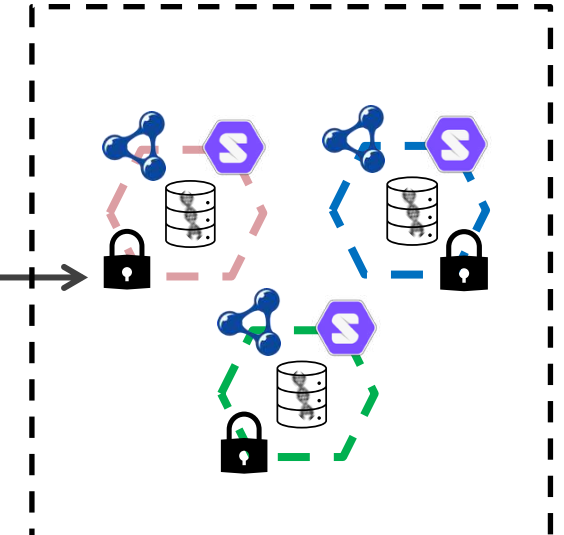
Data Storage



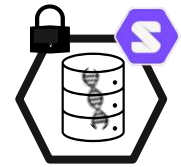
Representation



Privacy



Data Privacy Controls



Granular privacy controls can be implemented in Solid for genomic data.



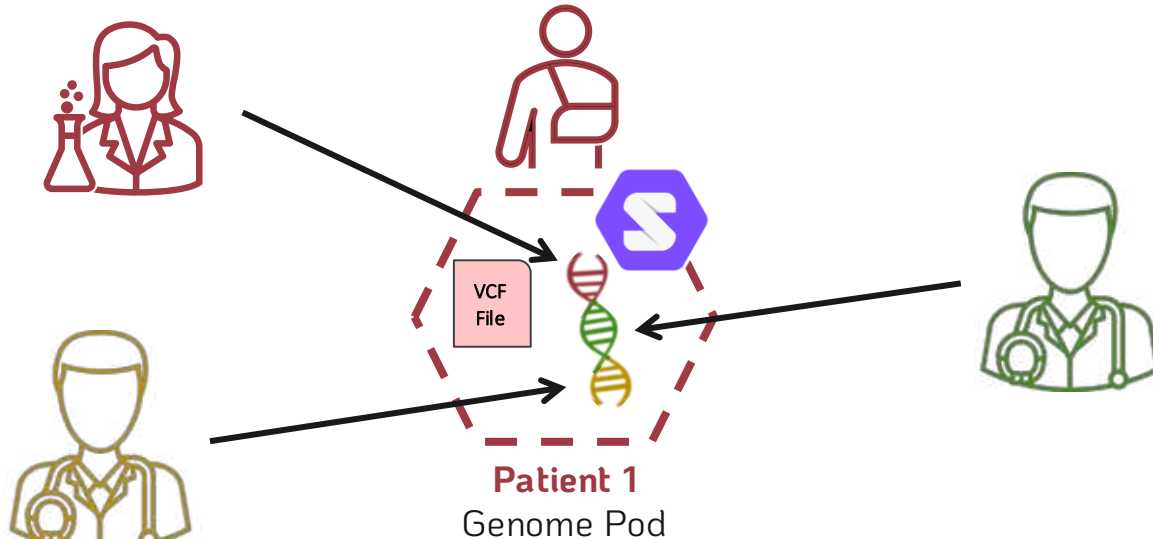
Web Access Control



Solid Protocol ACP³

How granular can we make the privacy controls?

Privacy controls at chromosome level, or even mutation level.



Chromosome is a semantic Class

Mutation is also a semantic Class,
above a triple but below a Chromosome

Querying Genome Pods



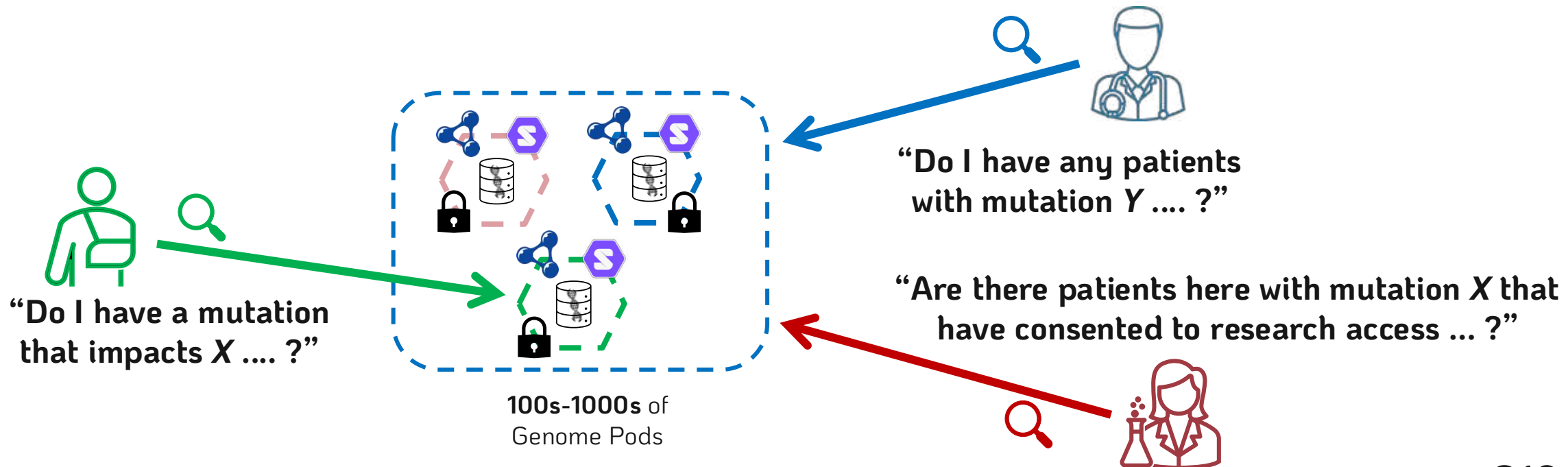
RDF genomic data pods can be queried.



Query Engine



Custom Comunica implementation/engine⁶



Querying Genome Pods (possibilities)



Single Pod Query

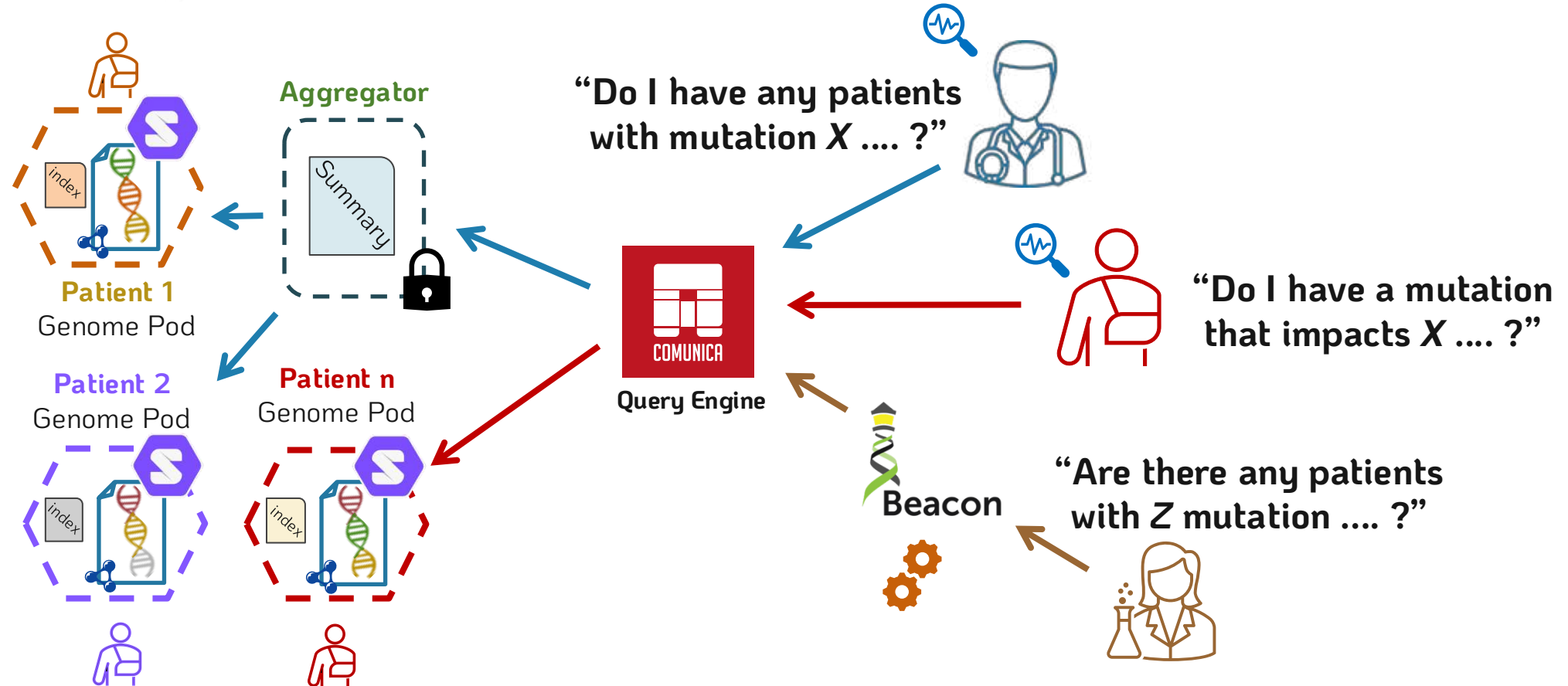
Utilizes an intra-pod index

Many Pod Query

First, utilizes an extra-pod aggregator
Then, utilizes the intra-pod indexes

Beacon API for Researcher Queries

Query is translated from Beacon API to Comunica
Then, the **Many Pod** query approach is followed



Implementation Specifics



Solid pod instances → Community Solid Server¹

Genomic Data → Publicly available *VCF* data²

RDF Conversion → HERO Genomics + SPHN ontology⁴

Data-linking → Actively exploring ontologies / methodologies

Data Querying → Custom instance of Comunica⁶